LAB 6

Code and Branch Coverage Testing

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Repository	L6_L7_13_pet_walking_service

Lab handout / execution report

Testing the Pet Walking Service module with pytest and pytest-cov.

Objective

To write and execute automated tests for the Pet Walking Service module, then measure statement (line) coverage and branch coverage using pytest-cov.

1. Overview

The provided module contains two functions: **process_walking_routes()**, which calculates walking-route revenue and handles route/session conditions, and **inspect_gps_collars()**, which checks collar GPS signal logs for consecutive weak readings.

2. Testing Procedure

1. Install pytest and pytest-cov in the Python environment.

2. Run the starter test to confirm that the test environment works.

3. Add test cases for route validation, session types, pet counts, medication, weekend transport, aggression alerts, incident auditing, and GPS signal conditions.

4. Execute the full test suite with pytest.

5. Run pytest-cov with branch coverage enabled to measure statements and branches.

3. Commands Used

Run the test suite:

```
python -m pytest -v
```

Measure line and branch coverage:

```
python -m pytest --cov=pet_walking_service --cov-branch --cov-report=term-missing
```

4. Test Coverage Areas

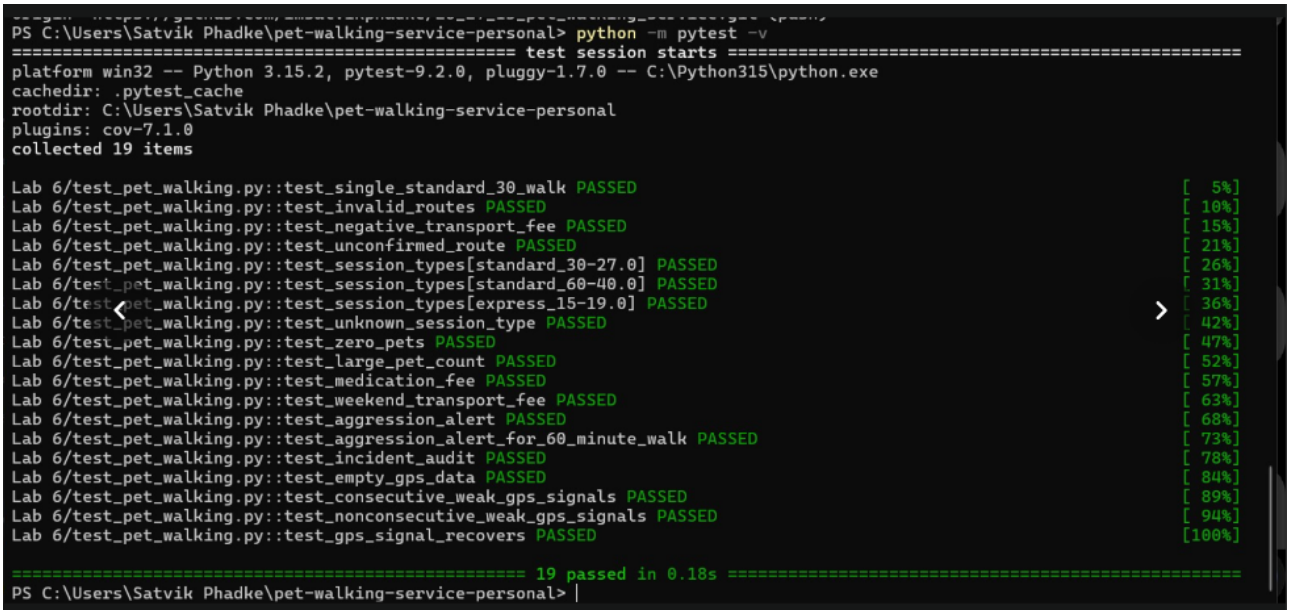
Area	Tested conditions
Input validation	Invalid routes input and negative transport fee
Route handling	Unconfirmed routes and unknown session type
Session pricing	standard_30, standard_60, and express_15
Walk conditions	Zero pets, more than three pets, medication, and aggression alert
Weekend and audit	Weekend transport fee and incident audit status
GPS collar inspection	Empty data, consecutive weak signals, nonconsecutive weak signals, and signal recovery

5. Results

The final test run in the personal repository completed successfully: **19 tests passed**. The coverage report shows 57 statements, 0 missed statements, 40 branches, and 0 partial branches.

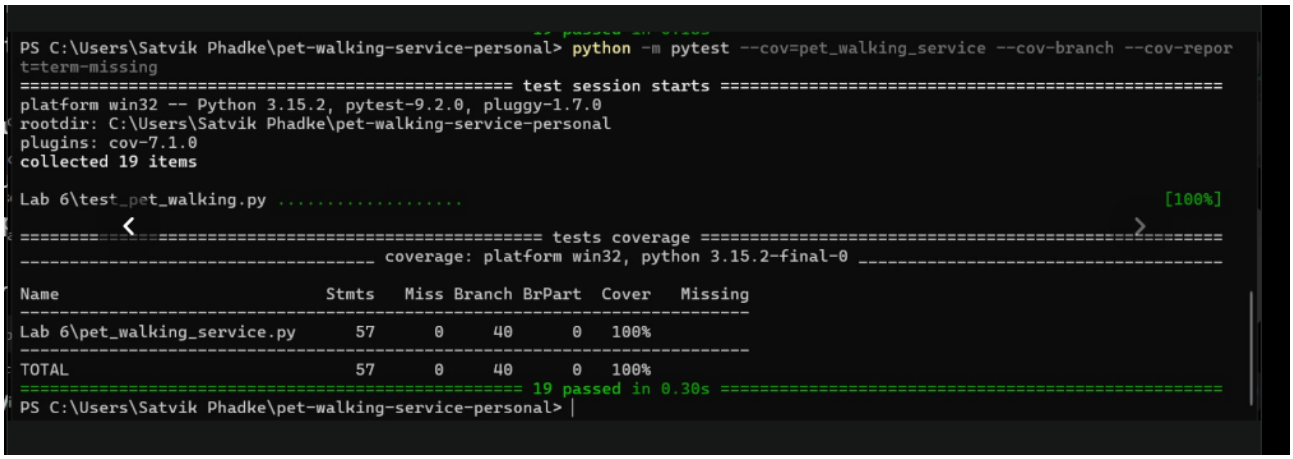
Metric	Result
Tests passed	19 / 19
Statements covered	57 / 57
Line coverage	100%
Branch coverage	100%
Missing statements	0
Partial branches	0

Evidence 1 — Pytest test execution



Screenshot showing all 19 tests passing in the personal clone.

6. Coverage Report Evidence



Screenshot showing 100% coverage: 57 statements covered and all 40 branches covered.

7. Conclusion

The test suite exercised all measured statements and branch parts in the Pet Walking Service module. The final run completed with 19 passing tests and 100% line and branch coverage according to pytest-cov. The screenshots included in this report document the test execution and coverage results.

Note: This report is based on the provided screenshots and the described test workflow. Add any instructor-specific requirements from the official lab handout if applicable.